MATH 5080 Class Notes

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1 The real number system

1.1 Natural numbers, integers, and rationals

Definition 1.1 (Natural numbers)**.** We define the set $\{1, 2, 3, 4, \ldots\}$ as the set of **natural numbers**, denoted $\mathbb{N}$.

Remark. $\mathbb{N}$ is closed under addition and multiplication, but not subtraction.

Definition 1.2 (Integers)**.** We define the set $\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ to be the set of **integers** denoted $\mathbb{Z}$.

Remark. Integers are closed under subtraction, but not division.

Definition 1.3 (Rationals)**.** We define the set

$$\left\{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\right\}$$

to be the set of **rational numbers**, denoted $\mathbb{Q}$.

Remark. $\mathbb{Q}$ has holes. For example, it can be shown that $x^2 = 2$ does not have a solution in $\mathbb{Q}$.

Proof. Suppose otherwise, then

$$\exists p, q \in \mathbb{Q}, q \neq 0, s.t. \frac{p^2}{q^2} = 2$$

WLOG, we can pick p, q such that they are coprime i.e. $gcd(p, q) = 1$.

$$\begin{aligned}
\frac{p^2}{q^2} = 2 &\implies p^2 = 2q^2 \\
&\implies p = 2k \text{ for sume } k \\
&\implies (2k)^2 = 2q^2 \\
&\implies 2k^2 = q^2 \\
&\implies q = 2k' \text{ for some } k' \\
&\implies gcd(p, q) \geq 2 > 1
\end{aligned}$$

✓

Remark. $x^2 = p$ for any prime p does not have a solution in $\mathbb{Q}$.

Proof. Assume otherwise, then

$$\begin{aligned}
&\exists m, n \in \mathbb{Z}, n \neq 0, s.t. \frac{m^2}{n^2} = p, gcd(m, n) = 1 \\
\implies & m^2 = n^2 p \\
\implies & m = pk \text{ for some } k, \text{ since } p \text{ is prime} \\
\implies & p^2 k^2 = n^2 p \\
\implies & pk^2 = n^2 \\
\implies & n = pk' \text{ for some } k' \text{ since } p \text{ is prime} \\
\implies & gcd(m, n) \geq p > 1
\end{aligned}$$

✓

Definition 1.4 (Real numbers)**.** The **real numbers**, denoted $\mathbb{R}$ is a set that

(i) is an ordered field
(ii) contains $\mathbb{Q}$ as an sub-ordered field
(iii) satisfies the completness axiom, or the least upper bound property

Remark. Such a set is unique under isomorphisms.

1.2 Ordered Field

Definition 1.5 (Field)**.** : A field F is a set with two binary operations

$$+ : F \times F \to F,\ (x, y) \mapsto x + y$$

$$\cdot : F \times F \to F,\ (x, y) \mapsto x \cdot y$$

that satisfy these properties:

- (A1) additive closure: $x, y \in F, x + y \in F$
- (A2) additive commutativity: for all $x, y \in F$, $x + y = y + x$
- (A3) additive associativity: for all $x, y, z \in F$, $x + (y + z) = (x + y) + z$
- (A4) existence of additive identity or neutral element: there is $0 \in F$ such that $x + 0 = x$ for all $x \in F$
- (A5) existence of additive inverse: for all $x \in F$ there is y such that $x + y = 0$
- (M1) multiplicative closure: $x, y \in F, x \cdot y \in F$
- (M2) multiplicative commutativity: for all $x, y \in F$, $x \cdot y = y \cdot x$
- (M3) multiplicative associativity: for all $x, y, z \in F$, $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- (M4) existence of multiplicative identity or neutral element: there is $1 \in F, 1 \neq 0$ such that $x \cdot 1 = 1 \cdot x = x$ for all x
- (M5) existence of multiplicative inverse: for all $x \in F, x \neq 0$ there is y such that $x \cdot y = 1$
- (D) distributivity: for all $x, y, z \in F$, $(x + y) \cdot z = x \cdot z + y \cdot z$

Remark. Additive neutral element, 0, is unique.

Proof. Suppose otherwise, i.e. $\exists 0, 0' \in F$ both satisfying A4, then

$$\begin{aligned} 0 &= 0 + 0' \text{ by A4, with } 0' \text{ as identity} \\ &= 0' + 0 \text{ by commutativity} \\ &= 0' \text{ by A4, since 0 is additive identity} \end{aligned}$$

✓

Remark. Multiplicative identity, 1, is unique.

Proof. Suppose otherwise, i.e. $\exists 1, 1' \in F$ both satisfying A4, then

$$\begin{aligned} 1 &= 1 \cdot 1' \text{ by M4, with } 1' \text{ as identity} \\ &= 1' \cdot 1 \text{ by commutativity} \\ &= 1' \text{ by M4, since 1 is identity} \end{aligned}$$

✓

Remark. $\forall x \in F$, the additive inverse is unique.

Proof. Suppose otherwise, then $\exists y, y'$ s.t. $x + y = x + y' = 0$.

$$\begin{aligned} y &= y + 0 \\ &= y + (x + y') \\ &= (y + x) + y' \text{ by associativity} \\ &= (x + y) + y' \text{ by commutativity} \\ &= y' \end{aligned}$$

✓

Remark. $\forall x \in F, x \neq 0$, the multiplicative inverse is unique.

Proof. Suppose otherwise, then $\exists y, y'$ s.t. $x \cdot y = x \cdot y' = 1$.

$$\begin{aligned} y &= y \cdot 1 \\ &= y \cdot (x \cdot y') \\ &= (y \cdot x) \cdot y' \text{ by associativity} \\ &= (x \cdot y) \cdot y' \text{ by commutativity} \\ &= y' \end{aligned}$$

✓

Remark. Since the additive and multiplicative inverses are unique, we denote $-x$ and $\frac{1}{x}$, and we define subtraction as

$$x - y := x + (-y)$$

We define division as

$$\frac{x}{y} := x \cdot \left(\frac{1}{y}\right), y \neq 0$$

Proposition 1.6. Let $x, y, z \in F$, then

1. $x + y = y + z \implies y = z$
2. $-(-x) = x$
3. $xy = xz, x \neq 0 \implies y = z$
4. $\frac{1}{1/x} = x$ if $x \neq 0$
5. $0 \cdot x = 0$
6. $-0 = 0$
7. $(-1) \cdot x = -x$
8. $xy = 0 \Leftrightarrow x = 0$ or $y = 0$
9. $(-x)(-y) = xy$

Definition 1.7 (Ordered field)**.** A field F is an **ordered field** if it can be equipped with a relation $<$ so that

1. $\forall x, y \in F$, exactly one of $x < y, x = y, y < x$ holds
2. $\forall x, y, z, \in F, x < y, y < z \implies x < z$
3. $\forall x, y, z \in F,\ x < y \implies x + z < y + z$
4. $\forall x, y, \in F,\ 0 < x, 0 < y \implies 0 < xy$

Proposition 1.8. Let F be an ordered field and $x, y, z, w \in F$. Then

1. $x > y, z > 0 \implies xz > yz$
2. $x > y, z < 0 \implies xz < yz$
3. $x > y \Leftrightarrow -x < -y$
4. $x < y, z < w \implies x + z < y + w$
5. $0 < x < y \implies 0 < \frac{1}{y} < \frac{1}{x}$
6. $\forall x \neq 0, x^2 > 0$. In particular, $1 > 0$

Definition 1.9 (Absolute value)**.** For all $x \in \mathbb{R}$, we define its **absolute value** $|x|$ as

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0, \end{cases}$$

Proposition 1.10. Let $x, y \in \mathbb{R}, a \geq 0$, then

1. $|x| \geq 0$
2. $|x| \leq a \Leftrightarrow -a \leq x \leq a$
3. $|xy| = |x| \cdot |y|$
4. (triangle inequality) $|x + y| \leq |x| + |y|$

Proposition 1.11. Let $x, y \in \mathbb{R}$ such that $x \leq y + \epsilon, \forall \epsilon > 0$, then $x \leq y$.

Proof. We prove by contrapositive. Assume that $x < y$, WTS that $\exists \epsilon > 0, y + \epsilon < x$.

Pick

$$\epsilon := \frac{x - y}{2}$$

Then

$$\begin{aligned} y + \epsilon &= y + \frac{x - y}{2} \\ &= \frac{x + y}{2} \text{ by (D), (A2), (A3)} \\ &< \frac{x + x}{2} \text{ by (O3)} \\ &< x \end{aligned}$$

✓

1.3 Completeness Axiom / LUB property

Definition 1.12 (Bounded)**.** Let $S \subset \mathbb{R}$, we say S is **bounded from above** if $\exists m \in \mathbb{R}$ s.t. $s \leq m, \forall s \in S$. Such an m is called an **upper bound** of S.

Example. $[0, 1] := \{x \in \mathbb{R}, 0 \leq x \leq 1\}$ has upper bounds 1, 1.5, 2, etc.

Example. $\mathbb{N}$ is not bounded from above.

Definition 1.13 (Maximum)**.** If an upper bound m of S is an element of S, then m is called the **maximum** of S, denoted

$$\max S$$

Example. $S = [0, 1\}$ does not have a maximum, i.e. $\max S$ DNE.

Proof. $\forall m \in S$, take $\epsilon = \frac{1}{2}(1 - m)$. Then $m + \epsilon < 1, m + \epsilon > m$. Hence m is not an UB. Hence any m cannot be UB. ✓

Definition 1.14 (Supremum / least upper bound)**.** Let $S \in \mathbb{R}$ be bounded from above. Then $m \in \mathbb{R}$ is called the **supremum**, or **least upper bound** if

1. m is an upper bound of S
2. $\forall m' \in \mathbb{R}$, if $m' < m$ then m' is not an upper bound of S.

Remark. $\sup[0, 1\} = 1$.

Proof. Suppose $\sup[0, 1\} \neq 1$. Since 1 is an upper bound and (1.14.1) holds, then 1.14.2 must fail. Suppose $m < 1$ is an upper bound, but $\frac{1}{2}(m + 1) \in S, \frac{1}{2}(m + 1) > m$. ✓

Remarks.

1. $\sup S$ is unique.
2. $\sup S$ may not be an element of S.
3. If $\max S$ exists, then $\max S = \sum S$

Example. $S = \emptyset$, then any $x \in \mathbb{R}$ is an upper bound and lower bound of S, but $\max S, \min S, \sup S, \inf S$ do not exist.

Theorem 1.15 (Completeness Axiom)**.** If $S \subset \mathbb{R}, S \neq \emptyset$, and S bounded from above, then $\sup S$ exists.

Remark. The completeness axiom doe not hold in $\mathbb{Q}$.

Proof. Take $S = \{x \in \mathbb{Q} : x \geq 0, x^2 < 2\}$.
We claim that any $m \in \mathbb{Q}$ upper-bounds S cannot be its supremum. ✓

Theorem 1.16 (Archimedean Property)**.** $\mathbb{N}$ is unbounded from above in $\mathbb{R}$.

Proof. Suppose $\mathbb{N}$ is bounded from above by $m \in \mathbb{R}$.

$$\begin{aligned}
&\implies \exists \sup \mathbb{N} \in \mathbb{R} \text{ by CA} \\
&\implies m' := \sup \mathbb{N} \text{ is LUB} \\
&\implies m' - 1 \text{ is not an upper bound} \\
&\implies \exists n \in \mathbb{N} \text{ s.t. } m' - 1 < n \\
&\implies m' < n + 1 \in \mathbb{N}
\end{aligned}$$

This contradicts with the assumption that m' is the supremum. ✓

Result 1.17 (Application of CA)**.** $x^2 = 2, x > 0$ has a unique solution in $\mathbb{R}$, called $\sqrt{2}$

Proof. Let $S := \{x \in \mathbb{R}, x > 0, x^2 < 2\}$. We claim that

- $S \neq \emptyset$. Take $1, 1 > 0, 1^2 < 2$. Check
- S is bounded from above. Take 2. Suppose $\exists x \in S, x > 0$ then $x^2 > 2$.

Hence by CA, S has a supremum in $\mathbb{R}$, say m.

Claim: $m > 0, m^2 = 2$.

1. if $m \leq 0$, then contradiction. Hence $m > 0$ check.
2. To show $m^2 = 2$, it suffices to show that $m^2 > 2$ or $m^2 < 2$ are both false.
   (a) Assume $m^2 < 2$. Then we can find $n \in \mathbb{N}$ s.t. $\left(m + \frac{1}{n}\right)^2 < 2$. **For exercise**
   (b) Assume $m^2 > 2$. **For exercise**

✓

Definition 1.18 (Density)**.** $A \subset B$ is dense if $\forall x \in B, \forall \epsilon > 0, \exists y \in A$ s.t. $|x - y| < \epsilon$.

You can do ϵ-approximation for anythingh in B with something in A.

Theorem 1.19. We have

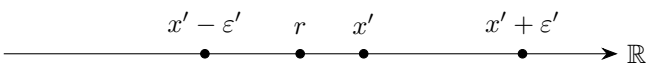
1. $\mathbb{Q}$ is dense in $\mathbb{R}$.
2. $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

Remark. $\mathbb{Q}$ dense in $\mathbb{R} \Leftrightarrow \forall a \in \mathbb{R}, b \in \mathbb{R}, a < b, \exists r \in \mathbb{Q}$ s.t. $a < r < b$.

Remark. $r \in \mathbb{Q}, r \neq 0 \implies \sqrt{2}r \in \mathbb{R} \setminus \mathbb{Q}$

Proof. Thm 1.19.

1. We first show that (1) $\implies$ (2), i.e. $\mathbb{Q}$ dense $\implies \setminus R \setminus \mathbb{Q}$ dense.
   $\forall x \in \mathbb{R}, \forall \epsilon > 0$. WTS $\exists y \in \mathbb{R} \setminus \mathbb{Q}$ s.t. $|x - y| < \epsilon$.
   Let $x' = \frac{x}{\sqrt{2}}$, $\epsilon' = \frac{\epsilon}{\sqrt{2}}$. Since $\mathbb{Q}$ dense in $\mathbb{R}$, $\exists r \in \mathbb{Q}$ s.t. $r \neq 0, |x' - r| < \epsilon'$. (If $r = 0$, then we can pick a smaller window $\epsilon'' = \frac{r - x'}{2}$.)

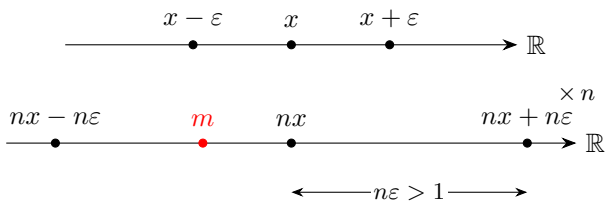


   Then, we take $y = \sqrt{2}r$.

$$\begin{aligned}
|x - \sqrt{2}r| &= \sqrt{2}|x' - r| \\
&< \sqrt{2}\epsilon' \\
&= \epsilon
\end{aligned}$$

2. Now we show (1), i.e. $\mathbb{Q}$ is dense in $\mathbb{R}$. I.e. WTS $\forall x \in \mathbb{R}, \forall \varepsilon > 0\ \exists y \in \mathbb{Q}$ s.t. $|x - y| < \varepsilon$.

   By Archimedean property, $\exists n \in \mathbb{N}$s.t.$n\varepsilon > 1$



   Also $\forall y \in \mathbb{R}, \exists m \in \mathbb{Z}$s.t.$|y - m| < 1$ Hence take $y = nx$

$$|nx - m| < 1 < n\varepsilon$$
$$\implies \left| x - \frac{m}{n} \right| < \frac{n\varepsilon}{n} = \varepsilon$$

✓

2 Cardinality

Definition 2.1 (Finite)**.** A set $S \neq \emptyset$ is **finite** if $\exists n \in \mathbb{N}$ and a bijection $f : S \to \{1, 2, \dots n\}$. Otherwise S is **infinite**.

Definition 2.2 (Countability)**.**

1. A set S is **countably infinite** if there is a bijection $f : S \to \mathbb{N}$
2. A set S is **countable** if it is empty or countably infinite
3. A set S is **uncountable** if it is not countable.

Proposition 2.3.

1. Any subset of a countable set is countable.
2. A countable union of countable sets is countable.
3. (Follows from (2)) $\mathbb{Q}$ is countable. **Exercise**
4. $\mathbb{R} \setminus \mathbb{Q}$ is uncountable $\Leftrightarrow$ $\mathbb{R}$ is uncountable. **Exercise**
5. Union of a countable set and an uncountable set is uncountable.

3 Metric spaces and point-set topology

Definition 3.1 (Metric space)**.** A **metric space** (X, d) is a set X of elements (called points) equipped with a metric or function $d : X \times X \to [0, \infty)$ such that

1. $\forall x, y \in X, d(x, y) \geq 0$. i.e. The image of d is in $\mathbb{R}$ and non-negative
2. $d(x, y) = 0 \Leftrightarrow x = y$
3. $\forall x, y \in X, d(x, y) = d(y, x)$
4. $\forall x, y, z, \in X,\ d(x, z) \leq d(x, y) + d(y, z)$

Definition 3.2 (Discrete metric)**.** The **discrete metric** d is defined such that $\forall x, y$

$$d(x, y) = \mathbf{1}[x \neq y] = \begin{cases} 0, & \text{if } x = y \\ 1, & \text{if } x \neq y \end{cases}$$

Example. The followign are metric spaces

- $X = \mathbb{R}, d(x, y) = |x - y|$
- $X = \mathbb{R}^n := \{(x_1, x_2, \dots x_n) : x_i \in \mathbb{R}\}$, $d_1(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i|^2 \right)^{\frac{1}{2}}$, i.e. Euclidean distance
- $X = \mathbb{R}^n$, $d_2(x, y) = \max_i |x_i - y_i|$
- X is the set of continuous functions on $[a, b]$, and $d(f, g) = \sup_{x \in [a,b]} |f(x) - g(x)|$
- Any $X \neq \emptyset$ with the discrete metric.

Definition 3.3 (Neighborhood)**.** Let (X, d) be a metric space. Let $x \in X, \varepsilon > 0$. The **ε-neighborhood** of x is

$$N(x, \varepsilon) = B_\varepsilon(x) := \{y \in X, d(x, y) < \varepsilon\}$$

A **deleted neighborhood** is

$$\begin{aligned} N^*(x, \varepsilon) : &= \{y \in X, 0 < d(x, y) < \varepsilon\} \\ &= N(x, \varepsilon) \setminus \{x\} \end{aligned}$$

Definition 3.4 (Interior)**.** Let (X, d) be a metric space, $S \subset X$. A point $x \in S$ is an **interior point** of S if

$$\exists \varepsilon > 0 \text{ s.t. } N(x, \varepsilon) \subset S$$

I.e. If an ε-neighborhood around x is in S, then x is an interior point.

The set of all interior points of S is called the **interior** of S, denoted $\operatorname{int} S$

$$\operatorname{int} S := \{x \in S : \exists \varepsilon > 0 \text{ s.t. } N(x, \varepsilon) \subset S\}$$

Definition 3.5 (Boundary)**.** Let (X, d) be a metric space, $S \subset X$. A point $x \in S$ is a boundary point of S if $\forall \varepsilon > 0$,

$$\begin{aligned} N(x, \varepsilon) \cap S &\neq \emptyset \\ N(x, \varepsilon) \cap S^c &\neq \emptyset \end{aligned}$$

The set of boundary pints of S is $\operatorname{bd} S$

$$\begin{aligned} \operatorname{bd} S : &= \{x \in X : \forall \varepsilon > 0, N(x, \varepsilon) \cap S \neq \emptyset, N(x, \varepsilon) \cap S^c \neq \emptyset\} \\ &= (\operatorname{int} S \cup \operatorname{int} S^c)^c \end{aligned}$$

Example. $\operatorname{int} \mathbb{N} = \emptyset$,

Proof. Since $\forall x \in \mathbb{N},\ \nexists \varepsilon > 0$ s.t. $N(x, \varepsilon) < \mathbb{N}$ hence $x \notin \operatorname{int} \mathbb{N}$. ✓

Example. $\operatorname{bd} \mathbb{N} = \mathbb{N}$.

Proof. Exercise ✓

Definition 3.6 (Closure)**.** The **closure** of S, denoted $\overline{S}$, is defined as

$$\overline{S} := S \cup \operatorname{bd} S$$

Definition 3.7 (Openess and closedness)**.**

1. $S \subset X$ is called **open** if $S = \operatorname{int} S$
2. $S \subset X$ is called **closed** if S^c is open.

Remark. A point cannot be both an interior point and a boundary point.

Remark. A set can be both open and closed. For example

- $\mathbb{R}$ is both open and closed.
- $\emptyset$ is both open and closed.
- $\mathbb{N}$ is not open, but closed.

Example. $S = (1, 3)$ is open and not closed.

Example. Let (X, σ) be a metric space where σ is the discrete metric. Take $x_0 \in X$, $S = \{x_0\}$ is open.

To see why, let $\varepsilon = \frac{1}{2}$.

$$\begin{aligned} & N\left(x_0, \frac{1}{2}\right) = \{x_0\} \subset S \\ \implies & x_0 \in \operatorname{int} S \\ \implies & \operatorname{int} S = S \\ \implies & S \text{ open} \end{aligned}$$

Hence any point in a set is interior, and any set is open and closed.

Proposition 3.8.

1. S open $\Leftrightarrow \operatorname{bd} S \subset S^c$
2. S closed $\Leftrightarrow \operatorname{bd} S \subset S \Leftrightarrow S = \overline{S} := S \cup \operatorname{bd} S$
3. $\overline{S}$ is closed
4. $\operatorname{int} S$ is open $\implies \operatorname{int}(\operatorname{int}(S)) = \operatorname{int}(S)$

Proof. S **closed** $\Leftrightarrow \operatorname{bd} S \subset S$**.**
By definition, S closed $\Leftrightarrow S^c$ open. Also by definition, $\operatorname{bd} S = \operatorname{bd} S^c$.

$$\begin{aligned} \operatorname{bd} S \subset S &\Leftrightarrow \operatorname{bd} S^c \subset S && \text{by definition} \\ &\Leftrightarrow \operatorname{bd} S^c \cap S^c = \emptyset \\ &\Leftrightarrow \forall x \in S^c, x \notin \operatorname{bd} S^c \\ &\Leftrightarrow \forall x \in S^c, x \in \operatorname{int} S^c \\ &\Leftrightarrow S^c = \operatorname{int} S^c \\ &\Leftrightarrow S^c \text{ open} \\ &\Leftrightarrow S \text{ closed} \end{aligned}$$

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Proposition 3.9.

1. The union of open sets is open.
2. The intersection of finitely many open sets is open.

Add in analogous propositions about closed sets.

Example. $S_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$ is open for all $n \in \mathbb{N}$. $\bigcap_{n=1}^{\infty} S_n = \{0\}$ is closed.

To see why the intersection is closed, $\forall x > 0$, by AP, $\exists n \in \mathbb{N}$ s.t. $0 < \frac{1}{n} < x$. Hence $x \notin S_n$ and $S \notin \bigcap S_n$.

At the same time, $0 \in S_n, \forall n \in \mathbb{N}$.

Since $0 \notin \operatorname{int}\{0\}$, $\{0\}$ is not open.

Proof of 3.9.2. WTS intersection of finitely many open sets is open.

Let $S_1, \dots S_n$ be open sets, $S = \bigcap_{k=1}^{n} S_k$. WTS $\forall x S, \exists \varepsilon \in \mathbb{R}, \varepsilon > 0$ s.t. $N(x, \varepsilon) \subset S$. Note that

$$N(x, \varepsilon) \subset S \Leftrightarrow N(x, \varepsilon) \subset S_i, \forall i = 1, \dots, n$$

Since $x \in S_i$ and S_i is open, $\exists \varepsilon_i > 0$ s.t. $N(x, \varepsilon_i) \subset S_i$. Let $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\}$, and we know such ε exists because a sequence of finitely many numbers has a minimum. ✓

Definition 3.10 (Accumulation points and isolated points)**.**

1. Let $S \subset X$, a point $x \in X$ is an **accumulation point** of S if $\forall \varepsilon > 0$,

   $$N^*(x, \varepsilon) \cap S \neq \emptyset$$

   The set of all accumulation points of S is denoted S'.
2. If $x \in S \setminus S'$, then x is an **isolated point** of S.

Example. $S = \{0\} \cup (1, 2)$, $S' = [1, 2]$

Example. $S = \mathbb{Q} \cap (1, 2)$, $S' = [1, 2]$

Proposition 3.11.

1. $\overline{S} = S \cup S'$
2. S close $\Leftrightarrow S' \subset S$
3. $\overline{S}$ is the smallest closed set that contains S.

Proof of 3.11.1. WTS $S \cup \operatorname{bd} S = S \cup S'$.

1. We first show that $LHS \subset RHS$. $\forall x \in S \cup \operatorname{bd} S$, $x \in S \cup S'$.

   If $x \in S$, done.

   If $x \in \operatorname{bd} S$, $x \notin S$ then $x \in \operatorname{bd} S \cap S^c$. WTS $x \in S'$, i.e. $\forall \varepsilon > 0, N^*(x, \varepsilon) \neq \emptyset$.

   Since $x \in \operatorname{bd} S$, we know $N(x, \varepsilon) \cap S \neq \emptyset$. Since $x \in S^c$, we know $\exists y \in N(x, \varepsilon) \cap S, y \neq x$. Hence $y \in N^*(x, \varepsilon) \cap S$, and $N^*(x, \varepsilon) \neq \emptyset$.
2. We then show $RHS \subset LHS$, i.e. $\forall x \in S \cup S', x \in S \cup \operatorname{bd} S$.

   If $x \in S'$, done.

   If $x \in S', x \notin S$, $x \in S' \cap S^c$, WTS $x \in \operatorname{bd} S$ **Exercise**.

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